MATH 350-2 Advanced Calculus

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Outline

1. Real Analysis Lecture 14
   - Complete metric spaces
   - Metric subspaces

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Complete spaces

Definition

A metric space (M, d) is called **complete** if every Cauchy sequence in M converges to a point in M.

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Consider a space with a single point

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- every Cauchy sequence in M is convergent

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- every Cauchy sequence in M is convergent
- M is complete

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Euclidean space is complete

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- if we fill in all the missing stuff, we get $\mathbb{R}$ back!

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- therefore the space M is NOT complete

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$$d_S : S \times S \to \mathbb{R}, \quad d_S(x, y) = d(x, y).$$

The set (S, d_S) is a metric space, called a **metric subspace** of M.

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- the set $\mathbb{Q}$ with the Euclidean metric is a metric subspace of M
- the set $\mathbb{R}$ with the taxicab metric is not a metric subspace of M

Open subsets of subspace

Theorem

Let (S, d_S) be a subspace of the metric space (M, d). Then $U \subseteq S$ is open in the subspace if and only if there exists an open subset $V \subseteq M$ satisfying $U = V \cap S$.

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The **completion** of a metric space M is the space $\overline{M}$ of equivalence classes of Cauchy sequences in M with the equivalence relation that $\{x_n\} \equiv \{y_n\}$ if and only if $d(x_n, y_n) \to 0$.

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Metric on $\overline{M}$:

$$\overline{d}(\{x_n\}, \{y_n\}) = \lim_{n\to\infty} d(x_n, y_n).$$

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If M is already complete, then $M = \overline{M}$.

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Let (M, d) be a metric space and $S \subseteq M$ compact.

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To show that (S, d) is complete, we must show that every Cauchy sequence in S converges to a value in S.

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If the set X of all values of $\{x_n\}$ is finite, then the Cauchy condition implies $\{x_n\}$ converges.

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TIME OUT!



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Lemma (Budget Bolzano-Weierstrass)

Let (M, d) be a metric space and let $S \subseteq M$ be compact.

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Let (M, d) be a metric space and let $S \subseteq M$ be compact. Then for any infinite subset $X \subseteq S$, the set S has an accumulation point of X.

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To see this, suppose otherwise.
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The open cover $\{U_i : i \in I\}$ with $I = S$ and $U_i = B_S(i; r_i)$ of S has no finite subcover.

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The open cover $\{U_i : i \in I\}$ with $I = S$ and $U_i = B_S(i; r_i)$ of S has no finite subcover.
This is a contradiction, so S has an accumulation point of X. □

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Let $\epsilon > 0$.

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Then $\exists N > 0$ such that $m, n \geq N$ implies $d(x_m, x_n) < \epsilon/2$.

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Let L be an accumulation point of X in S.
Let $\epsilon > 0$.
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Moreover, $B_S(L, \epsilon/2)$ contains infinitely many points of X.

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Hence it contains x_m for some $m \geq N$.

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Moreover, $B_S(L, \epsilon/2)$ contains infinitely many points of X.
Hence it contains x_m for some $m \geq N$.
It follows that for all $n \geq N$,

$$d(x_n, L) \leq d(x_n, x_m) + d(x_m, L) < \epsilon/2 + \epsilon/2 = \epsilon.$$

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Since $\epsilon > 0$ was arbitrary, this proves $\{x_n\}$ converges to L. □